

Section B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(a)			$\text{Fe}_2\text{O}_3 + 3\text{CO} \rightarrow 2\text{Fe} + 3\text{CO}_2$ (1) Any of the following for (1) - During this process the iron(III) ions (in Fe_2O_3) gain electrons (to produce iron); reduction is a process of electron gain - The oxidation number of iron is reduced from +3 to 0; a reduction in (positive) oxidation number is reduction - Carbon monoxide loses electrons; oxidation is a process of electron loss - The oxidation number of carbon is increased from +2 to +4; an increase in (positive) oxidation number is oxidation Fe_2O_3 loses oxygen and CO gains oxygen	2			2		
	(b)			350 tonnes of which 0.02 % is sulfur $\therefore \text{Mass of sulfur} = \frac{350 \times 0.02}{100} = 0.07 \text{ tonnes}$ (1) $\text{Mg} + \text{S} \rightarrow \text{MgS}$ 0.07 tonnes sulfur needs $\frac{24.3 \times 0.07}{32.1} = 0.0530 \text{ tonnes}$ $= 53.0 \text{ (kg)}$ (1) ecf possible	2			2	1	
	(c)			Cubic structure shows alternating different ions (1) Ions labelled as Mg^{2+} and S^{2-} (1)		2		2		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8.	(d)	(i)		White solid / precipitate (of magnesium hydroxide)		1		1		1
		(ii)		Number of mol of MgS = $\frac{0.224}{56.4} = 0.00397$ $\therefore$ 0.00397 mol of H ₂ S also produced (1) Volume of H ₂ S = 0.00397 $\times$ 24.0 = 0.095(3) dm ³ (1) = 95.(3) cm ³ (1) ecf possible Accept alternative method using pV = nRT	1	2		3	1	
	(e)			A colourless solution (1) The solubility of the group 2 hydroxides increases down the group (1)	1	1		2		1
	(f)			BaO + H ₂ O $\rightarrow$ Ba(OH) ₂ (1) pH > 7 (1)	2			2		
	(g)			Any two of following for (1) each - Barium is in group 2 and has two outer electrons - Too much energy is needed to remove a third electron - This necessitates removing an electron from a shell nearer to the nucleus (to produce a Ba³⁺ ion)	1	1		2		
				Question 8 total	9	7	0	16	2	2